DATABRICKS × LANDMAN OIL · UPSTREAM SUBSURFACE

# Subsurface Intelligence

From Landman Oil's raw well logs to a located, valued, governed drilling decision, on one platform, in one sitting.

**$40M**
risked NPV10 of the next well, found in data Landman already has

**3.6x**
lower error filling a missing sonic log than today's method

**1 sitting**
from raw logs to a located, priced, governed well

Locate → Read → Fill the sonic → Model in voxels → Value → Decide

Working demo on Databricks · Lakeflow · Unity Catalog · MLflow · Model Serving · AI Gateway · Spatial SQL · Genie · Apps

01 · LANDMAN OIL'S BET

# Landman Oil has told the market it will grow the barrel and cut its cost at the same time. Every next-well decision sits on both.

**~$10B**
a year back into oil and gas

**2.3–2.5**
mmboe/d targeted by 2030, US upstream heading toward 1.0

**$4–5B**
of structural cost to take out

**~$5.60**
per barrel lifting cost to defend as volumes grow

DATA IN SILOS

### Several SaaS tools, one decision

Logs, tops, OSDU records, economics and maps sit in separate applications with separate owners. Every next-well proposal is a chain of exports and hand-offs.

GAPS IN THE RECORD

### Wells drilled without a key curve

A missing sonic log blocks geomechanics and seismic ties until someone patches it by hand.

AI STUCK IN PILOTS

### 86% of energy AI never leaves the pilot

Assistants either cannot see the governed data or see too much, so partners and JV data become the blocker.

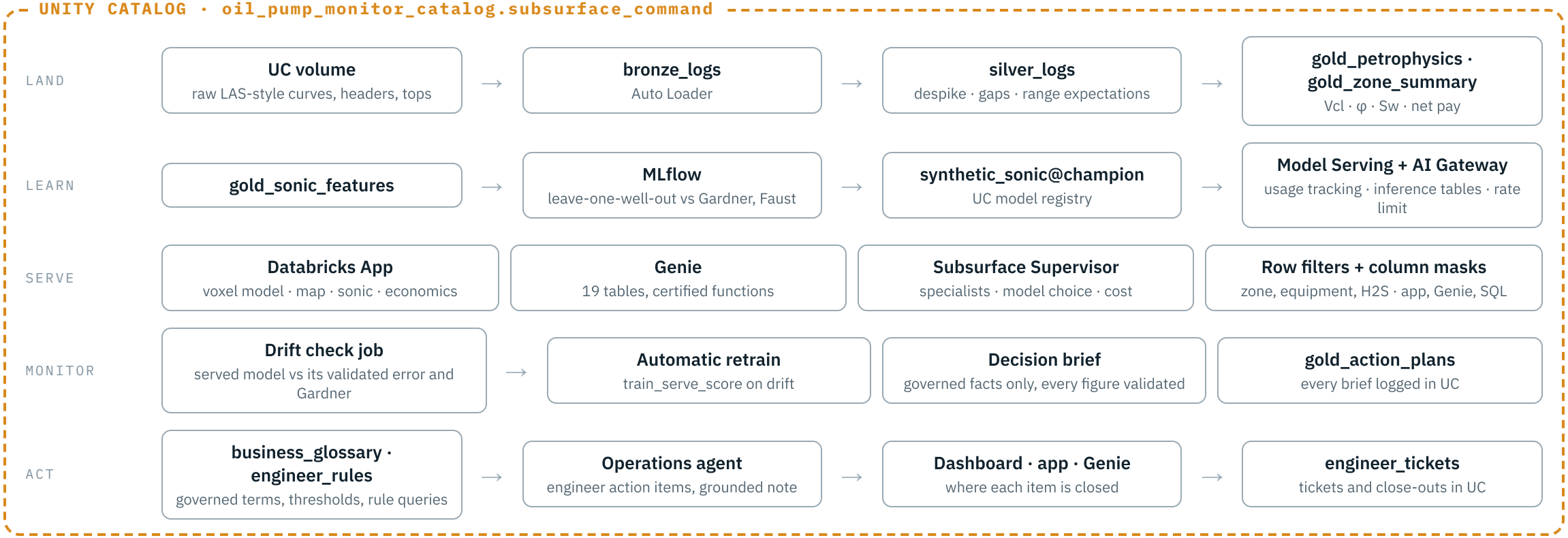
**Subsurface Intelligence puts Landman's logs, geomodel, economics and AI on one governed platform, so the next-well decision is one workflow, not five projects.**

# What it is worth to Landman Oil: a better well, found sooner, acted on the same morning

| LEVER | WHAT CHANGES | THE NUMBER |
|---|---|---|
| **1. Value per decision** | The geomodel finds and ranks a step-out in data Landman already has, and the certified NPV function prices it | **$103M** NPV10 unrisked, **$40M** risked, 34% IRR, $53 break-even on $58M capex demo |
| **2. Speed to decision** | Logs to a located, priced proposal in one sitting instead of a chain of hand-offs, so wells reach the program sooner on the way to 2.3–2.5 mmboe/d | Every **6 months** of delay costs **$1.9M** of the risked value (**$4.8M** unrisked), 4.7% of NPV at 10% demo. How long does a step-out proposal take at Landman today? ask |
| **3. No blocked decisions** | Missing curves are filled in seconds by a served model scored against the method engineers trust | **3.6x** lower error than calibrated Gardner on 13 held-out wells demo. How many Landman wells lack sonic? ask |
| **4. Safe to share** | Partners and AI assistants see only what they are entitled to, enforced once in Unity Catalog | JV partner sees **30 of 84** rows, interpretation masked, in the app, Genie and SQL demo |
| **5. One platform, many use cases** | The governed foundation built for this decision carries the next ten, instead of another stitched SaaS stack per use case | **60–70%** of an AI use case's cost is data integration; built once, the foundation serves **10+** use cases at **3–5x** lower cost, toward Landman's **$4–5B** cost target benchmark |
| **6. Act the same morning** | An operations agent runs the governed glossary rules over Unity Catalog data and hands the engineer prioritized action items, each linked to where it gets closed | **8** items on the 12,000 ppm H2S well, from an H2S audit **386 days** old to a wellhead spool with **20 days** of cover against a **60-day** lead demo |

**The cost side stays small: the sonic model is a scale-to-zero endpoint metered through AI Gateway, the pipeline runs on serverless, and the AI shows its cost per run. Value per decision is in the tens of millions; the platform cost per decision is not.**

# One pipeline, one registry, one governance layer




One Asset Bundle job, **subsurface-command-build**, lands the data, runs the pipeline, trains, registers, serves and scores. Rebuilding the whole stack is one command. Cost stays visible: serverless pipeline, a scale-to-zero endpoint with usage tracking and inference tables, and per-run token cost on the Supervisor. Every NPV you see, portfolio, step-out, Genie and Supervisor, comes from the certified **calculate_npv10** at today's WTI.

**The ontology is Unity Catalog itself:** glossary terms with owners and thresholds, tagged on the columns they govern; a PK/FK graph from tickets to rules to glossary terms and to wells; metric views, certified functions and governed tags. Change a threshold in the glossary and the agent, the dashboard and Genie follow, with no code.

04 · THE PATH, AND THE ASK

# Prove it on one Landman field with Landman's own data, then reuse the template

WEEKS 1–2

## Port

Deploy the bundle into Landman's workspace; land one field's LAS files, tops and headers, connected to the SaaS tools already in use.

WEEKS 3–4

## Score

Retrain the sonic model on Landman's wells; agree the bar it must clear against the current method.

WEEKS 5–8

## Decide

Voxel model on Landman's horizons and type curves; one step-out reviewed against the asset team's own pick.

AFTER

## Scale

Next field, next missing curve, JV partners under the same policies.

ASK 1 · SVP DIGITAL OPERATIONS

### Sponsor one field as the first production use case

**Measured on:** digital value delivered at scale and the contribution to the $4–5B structural cost target. **Success:** a step-out the asset team agrees is credible, priced in weeks rather than a quarter, and a template that takes the next field out of pilot.

ASK 2 · VP OF ENGINEERING

### A two-week working session with petrophysics and data

**Measured on:** platform reliability, reuse across use cases and governance. **Success:** Landman's logs, tops and OSDU records mapped to the pipeline, the model's acceptance bar agreed, partner policies live on Landman's tables, and engineers getting morning action items from Landman's own glossary rules.

APPENDIX · WHAT IS REAL, AND WHAT IS ILLUSTRATIVE

# Every screen says where its numbers come from

| ELEMENT | STATUS | DETAIL |
|---|---|---|
| Pipeline, model, endpoint, policies, Genie | Live | Running on Databricks; job, endpoint and policies verified |
| Sonic predictions | Live | Served model on every request; labelled Gardner fallback if cold |
| Voxel properties, step-out and its price | Live | Computed from the wells' logs and tops; priced by the certified UC function |
| Engineer action items, agent note, tickets | Live | Governed rules run over UC data each load; every figure in the note checked against the items; tickets written to UC |
| Glossary thresholds | Illustrative | Placeholder company standards (365-day audit, 180-day re-sample); Landman's HSE sets the real values |
| Well logs | Synthetic | Physically coupled rock model (porosity, shale, fluids); your logs replace them in the pilot |
| Structural framework | Illustrative | Dip, closure and fault are modeled, honoured at the tops; a real horizon model plugs in |
| OSDU analog panels | Offline | Fed by a partner ADME workspace that is not live; the panels say so |